

Name:

Roll No:

INSTITUTE OF BUSINESS ADMINISTRATION
Department of Computer Science, Karachi
Course: Introduction to Artificial Intelligence
Instructor: Dr. Syed Ali Raza
Final Exam

Duration: 3 hours

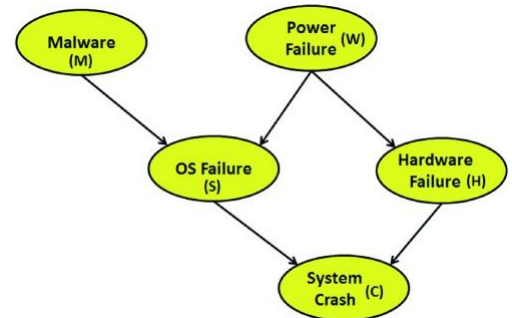
Total marks: 76

Instructions:

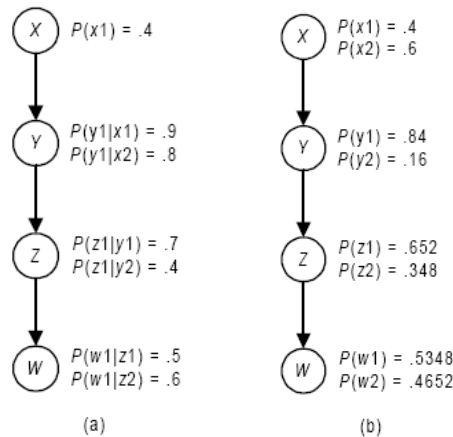
- In case of any ambiguity, you may make assumptions. But your assumption should not contradict any statement in the question paper. All the answers must be solved according to the sequence given in the question paper.

Short questions (36 marks, 4 each)

- For the Bayesian Network given here, state:
 - the joint probability equation based on the structure of the graph.
 - how many joint probabilities are required given each variable is binary?



- Suppose we have a singly connected Bayesian Network as shown in Figure (a) below. There are four variables: X, Y, Z, and W. The conditional probabilities are given in Figure (a), and the marginal probabilities of the four variables are given in Figure (b). Now, suppose we get evidence that w_2 is true, i.e., $P(w_2) = 1$. Find the posterior probabilities of $P^*(z_1)$ and $P^*(z_2)$.



- Assume that several facts of the following types are already defined using First Order Logic:

male(X)
female(X)
parent(X, Y)

For example, male(albert). female(elizabeth). parent(edward, albert). Etc.

Using ONLY the facts defined above, define rules in First Order Logic (FOL) or Prolog for the following relations:

mother(X,Y), daughter(X,Y), sibling(X,Y), grandMother(X,Y)

- 4) Validate the following two arguments using the propositional logic. Also, mention the rule of inference which applies.

Argument 1

Premise 1: I will not study discrete math or I will study English literature.

Premise 2: I will study discrete math or I will study databases.

Conclusion: Therefore, I will study databases or I will study English literature.

Argument 2

Premise 1: If it is raining, the ground is wet

Premise 2: If the ground is wet, the ground is slippery

Premise 3: It is raining.

Conclusion: The ground is not slippery.

- 5) Assume MyGame is in an unknown MDP where there are three states [A, B, C] and two actions [Stop, Go]. We are given the two samples generated from taking actions in the unknown MDP. Also, assume $\gamma = 1$ and $\alpha = 0.5$. We run Q-learning on these samples.

s	a	s'	r
A	Go	B	2
C	Stop	A	0

What are the estimates for the following Q-values as obtained by Q-learning? All Q-values are initialized to 0.

a) $Q(A, Go)$

b) $Q(C, Stop)$

Q-value update formula is given below,

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left(R(s, a, s') + \gamma \max_{a'} Q(s', a') - Q(s, a) \right)$$

- 6) Assume we have a robot for surveillance in a library. Its task is to move around the library and identify any wrongdoing (like, load music, playing games, etc.), and report the incident to the human operator. We want to apply reinforcement learning so the robot can perform its duties autonomously. Your task is to identify the elements of MDP (no need to apply reinforcement learning algorithm). It includes identifying the set of actions, the set of states, the rewards, and the transitions between states. Write down these elements of MDP for this task. You can make reasonable assumptions about the observations and the actions which the robot can perform.
- 7) Briefly answer the following questions about K-means clustering.
- Which method do we use to measure the performance of K-means? Explain the method using a simple example.
 - Write two post-processing steps for the K-means clustering.
- 8) Write the formula for measuring accuracy of a model in supervised learning in terms of true-positives (TP), false-positives (FP), true-negatives (TN) and false-negatives (FN). With a simple example explain what is the main limitation of using the accuracy measure?

- 9) Consider the following confusion matrix of a predictive model to detect spam e-mail. Find the values of:

- Precision
- Recall
- F-measure

		PREDICTION	
		Spam	Not Spam
ACTUAL	Spam	560	40
	Not Spam	50	50

Long Questions (40 marks, 8 marks each)

Note: the underlined text in the following questions highlights what is required in the answers.

- 10) The table below shows fictional statistics of student grades (pass and fail) in AI course. There are eight examples with three characteristics found important in determining the success of a student, namely Game Skills, Favorite Campus and Favorite Drink. Induce a Decision Tree (partially) using GINI Index as the measure of impurity.
- Which attribute is selected at root and what is the Gini Index of the root node selected?
 - Show the data distribution table for each child node of the root node.
 - Based on the data distribution table on each branch of the root, tell which node should be selected as a child on each branch of the root node according to the Gini Index (Hint: no need to actually calculate the Gini index) or if a decision can be reached on one or more branches.
 - Draw the decision tree formed after selecting the root node. Show if any branch or branches of the root node have a decision node (i.e., leaf node).

$$GINI(t) = 1 - \sum_j [p(j|t)]^2$$

Example	Game Skills	Favorite Campus	Favorite Drink	Grade
X1	Expert	Main	Next	Fail
X2	Novice	Main	Fizup	Fail
X3	Novice	City	Next	Pass
X4	Novice	City	Fizup	Fail
X5	Expert	City	Fizup	Fail
X6	Novice	City	Pakola	Pass
X7	Novice	Main	Next	Fail
X8	Expert	City	Next	Pass

- 11) For the data given here (9 instances of X and Y coordinates), apply the K-means algorithm for two iterations using K=3. Note: two iterations mean finding the new centroids two times. Use instance 3, 6, and 8 as the initial centroids. Use Manhattan distance as the distance measure.

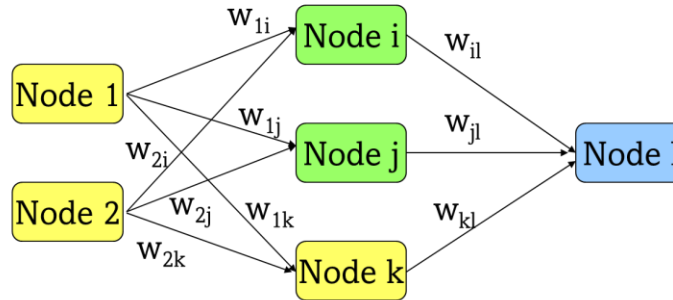
$$\text{Manhattan Distance} = |x_1 - x_2| + |y_1 - y_2|$$

- For each cluster formed after the first and the second iterations, write down the set of instances belonging to each cluster. For example, $C1=\{1,3,\dots\}$ and so on.
- Also, write down the new centroids obtained after the first and the second iteration.

Instance	X	Y
1	4	7
2	-1	3
3	2	-2
4	5	1
5	7	6
6	-5	-2
7	3	6
8	-3	-1
9	6	0

- 12) For the fully connected feed-forward neural network given here,
- find the output at node 1
 - calculate the error at node 1
 - calculate the updated values of weights w_{1i} , w_{2j} , and w_{kl} .
- The initial weight values are $w_{1i}=0.20$, $w_{1j}=0.10$, $w_{1k}=0.30$, $w_{2i}=-0.10$, $w_{2j}=-0.10$, $w_{2k}=0.20$, $w_{il}=0.10$, $w_{jl}=0.50$, $w_{kl}=0.20$.
 - Input is node1 = 0.4 and node2 = 0.6, and the target value is $T=0.65$. The learning rate (r) is 0.5. For all the calculations, use values till 5 decimal places.
 - Use the sigmoid function as the activation, $f(x) = 1 / (1 + e^{-x})$.
 - The formula for the error at node 1 is $\text{Error}(1) = (T - O_1) O_1 (1 - O_1)$ where O_1 is the output of node 1.
 - The formula to compute error at node i is $\text{Error}(i) = \text{Error}(1) w_{i1} O_1 (1 - O_1)$. Similarly, you can use formulas to compute errors at node j and k.
 - The formula to compute new weight between two nodes x and y is,
 - $w_{xy} = w_{xy} (\text{current}) + \Delta w_{xy}$

- where, $\Delta w_{xy} = r \times \text{Error}(y) * O_x$



13) Run one iteration of Evolutionary Algorithm to maximize the following

$$\text{function: } f(x,y) = x^2 + y^2 + z$$

Current generation with 5 solutions is given here.

	(x,y,z)
1	(1,-2,2)
2	(4,-2,3)
3	(-3,4,-1)
4	(5,2,1)
5	(-3,-5,3)

- Use Binary tournament scheme for parent selection and select 2 parents. The first tournament is between 1 and 5 and the second tournament is between 2 and 4. Show complete working on Answer sheet and show the final answer on answer sheet as:
 - Parent1: _____
 - Parent2: _____
- Perform crossover by taking x value of parent 1 and (y, z) value of parent 2 and vice versa to create 2 offsprings. Show complete working on Answer sheet and show the final answer on answer sheet as:
 - Offspring1: _____
 - Offspring2: _____
- Perform swap mutation on each offspring by swapping x and y gene values. Show complete working on Answer sheet and show the final answer on answer sheet as
 - Mutated Offspring1: _____
 - Mutated Offspring2: _____
- Use truncation as survivor selection to keep the population to 5 and state the 5 individuals remaining in the population. Show complete working on Answer sheet and show the final answer on answer sheet as a table on the right.

	(x,y,z)
1	
2	
3	
4	
5	

14) Apply Particle Swarm Optimization (PSO). Find the maximum of the function,

$$f(x) = -x^2 - 3x - 20$$

Use 4 particles with initial positions,

$$x_1 = -0.8, \quad x_2 = 1, \quad x_3 = -2, \quad x_4 = 1.5$$

The initial velocities of particles are the same as their initial positions (e.g. the initial velocity of particle 1 is -0.8). Moreover, inertial weight (w) is 1, C1 = 0.5 and C2=0.5.

Apply one complete iteration of PSO. Find the fitness of initial solutions, personal best solutions (pbest), global best (gbest), new velocities of each particle, updated positions of each particle. In addition, find the personal best solutions (pbest) and the gbest solution for the next iteration. No need to complete other steps for the second iteration. For the first iteration, the random numbers are $r_1 = 0.213$ and $r_2 = 0.876$. For all the calculations, use numbers to three decimal places.